



EXPERIMENT TO VERIFY NONLINEAR RELATIONSHIP BETWEEN CURRENT AND VOLTAGE IN A NON-OHMIC RESISTOR



AIM

Study the nonlinear relation between current and voltage in a non-Ohmic resistor.



LEARNING OBJECTIVES

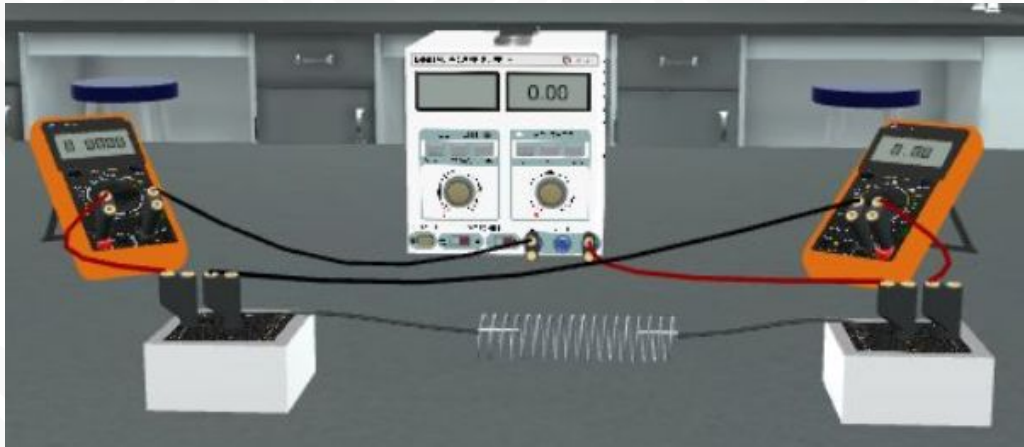
At the end of this lesson, you should be able to:

- Explain how resistance changes with temperature.
- Correlate the reciprocal relation between heat dissipated and current flow.
- Explain why Tungsten is used in heating application and lamps.
- Appreciate the intricacy of electric conductivity phenomena.



APPARATUS

- Voltmeter.
- Ammeter.
- DC power supply.
- A resistor made of Tungsten.
- A resistor made of Iron.





THEORY

Let us consider a particular sample of conducting material, apply a uniform potential difference across it, and measure the resulting current.

If we plot the results, the experimental points clearly fall along a straight line, which indicates that the ratio V/I is constant.

In this case, we say that the material obeys Ohm's law, which states that: a conducting device obeys Ohm's law if the resistance between any pair of points is independent of the magnitude and polarity of the applied potential difference.

Ohm's Law was formulated by Georg Simon Ohm, a German physicist, in 1827.



THEORY

- A material that obeys Ohm's law is said to be Ohmic.
- There are some elements that do not obey Ohm's law, where the current does not increase linearly with the applied voltage.
- Joule's law, which relates the heat (energy) generated in a conductor due to the flow of current, and how this affects the resistance of the conductor.
- Joule's law is often used to explain why certain elements do not follow Ohm's law under specific conditions.

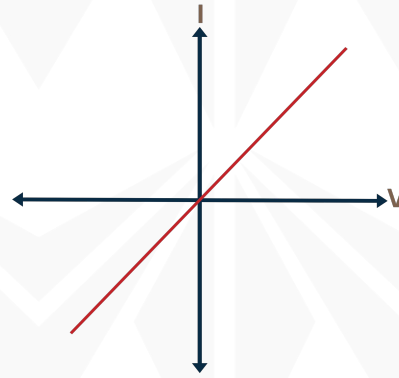


Fig. 1: Ohmic resistor

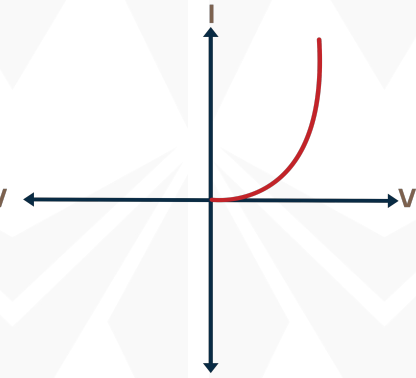


Fig. 2: Non-Ohmic resistor



THEORY

- Joule's law tells us that the power dissipated as heat in a conductor is directly proportional to the square of the current passing through it and also to its resistance.
- Joule's law can be expressed as: $P = I^2R$
- When current flows through a conductor, it encounters resistance, and this resistance results in the generation of heat (thermal energy).
- where P is the power dissipated as heat (in watts, W).
- I is the current flowing through the conductor (in amperes, A).
- R is the resistance of the conductor (in ohms, Ω).



THEORY

- As the current increases, the heat generated (P) increases as well, leading to a rise in the conductor's temperature.
- This rise in temperature can cause changes in the resistance (R) of the conductor, making it nonlinear in its response to voltage (V).
- This phenomenon is especially significant when current levels are high, and the conductor heats up considerably, leading to changes in its electrical properties and making it deviate from the linear relationship described by Ohm's law.
- The I - V (current-voltage) relationship becomes nonlinear, as shown in Figure (2) in your text, under these conditions.

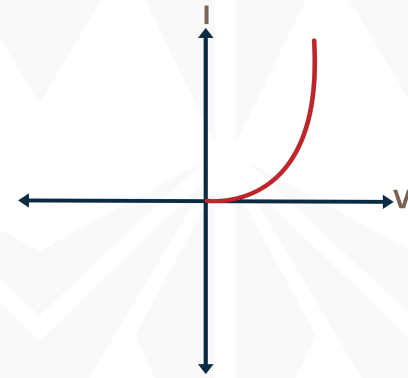


Fig. 2: Non-Ohmic resistor



MID-LESSON QUESTIONS

Question :

- (a) Can you provide an example of a real-world application where non-ohmic resistors are intentionally used.
- (b) Why are they preferred in such cases?



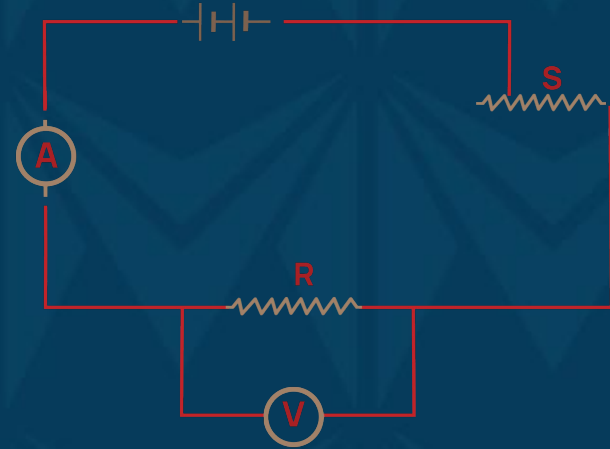
MID-LESSON ANSWERS

- (a) One common example of non-ohmic resistors in real-world applications is the use of THERMISTORS in temperature sensing circuits. Thermistors are semiconductor devices with resistance that changes drastically with temperature.
- (b) They are preferred in temperature sensing because their non-linear behaviour allows for precise measurement of temperature changes over a specific range. This sensitivity makes thermistors suitable for applications like temperature control in ovens, air conditioning systems, and automotive engine management, where precise temperature monitoring is essential.



PROCEDURE

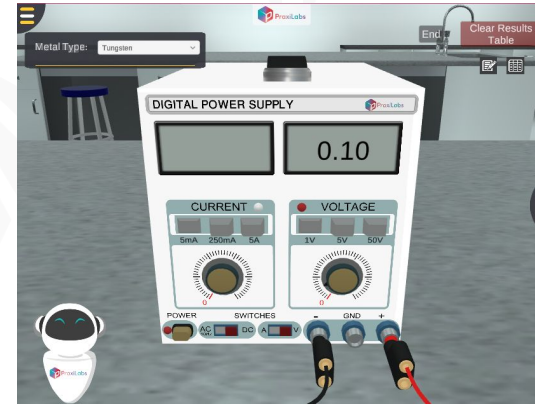
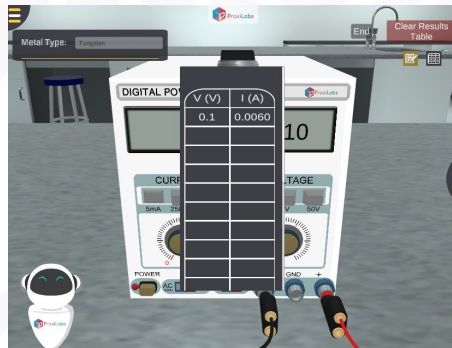
- Inspect the circuit connection.
- Select the material of the conductor from the drop list (tungsten or iron).





PROCEDURE

- Adjust the output voltage of the DC power supply 0.1 V (for example).
- Click the Record button to save both the readings of the voltmeter V and the current I .
- Increase the applied voltage to 0.3 V (for example).
- Click the Record button again to save the readings V and I .





PROCEDURE

- Repeat the previous steps by adjusting the applied voltage V several times and record the corresponding values of the current I .
- You should cover all the available voltage range (starting from 0.1 V to 50 V) using the three push buttons on the power supply above the voltage dial' knob.



PROCEDURE

- When you finish recording all the data, the experiment will be ended automatically.
- An Excel sheet will be downloaded containing the data you have saved.



DATA ANALYSIS

- The downloaded Excel sheet will look like:
- Plot a graph of I (A) vs V (V) which will be a curve representing the non-linear relation between I and V .
- Compute, I^2
- Compute P using IV
- Plot the graph of P (W) against I^2 (A^2). The graph should be linear. Illustrating Joules power Law

V (V)	I (A)
0.1	0.0060



WALK THROUGH VIDEO





FURTHER READING

Bueche F. and Hecht E. Schaum's Easy Outlines - College Physics Crash Course. 2000 McGraw-Hill, USA.

Hernández-Hall, C. A., & Wilson, J. D. (2014). Physics Laboratory Experiments. Cengage Learning.

Sang, D., Hewlett, C., Jones, G., Field, S., & Styles, D. (2020). Cambridge International AS & A Level Physics Practical Workbook. Cambridge University Press.

Serway, R.A. and Jewett. J.W. (2014). Physics for Scientists and Engineers with Modern Physics, Ninth Edition. Physical Sciences: Mary Finch.



FURTHER READING

Shukla, R. K., & Srivastava, A. (2006). Practical Physics. New Age International (P) Limited, Publishers.

Young, H. D. (2012). Sears & Zemansky's college physics. 9th ed. Pearson Education, Inc.

<http://hyperphysics.phy-astr.gsu.edu>

<http://www.physicstutorials.org>



THANK YOU